

MEMORANDUM

To: Mr. Sam
From: Duy Anh
Date: March 17, 2026
Subject: Why Portfolio Insurance Algorithms Sold Into the 1987 Crash

EXECUTIVE SUMMARY

Portfolio insurance was designed to protect institutional portfolios by reducing equity exposure as markets fell. On October 19, 1987, that portfolio-level discipline became a market-level vulnerability. As prices declined, many institutions received the same mechanical sell signal simultaneously and attempted to execute into rapidly thinning liquidity. What was rational for one portfolio became destabilizing in the aggregate. This memo examines how that failure unfolded and proposes three safeguards to prevent its recurrence.

The failure was not a coding error. It was structural: a procyclical strategy deployed at scale in a market that could no longer absorb the required trades. A strategy designed to limit downside at the portfolio level instead transmitted stress through the market when many institutions executed it simultaneously.

WHAT THE DATA SHOW

The minute-level S&P 500 futures data in Figure 1 indicates that the market did not experience a normal correction; it moved into disorderly liquidation. Prices were already trending lower on Friday, October 16, suggesting that de-risking had begun before the weekend. At the Monday open, however, selling became directional and sustained from the first minutes, with almost no countervailing buying pressure.

By 10:00 on October 19, rolling volatility had already crossed three times its pre-crash baseline. The worst 60-minute window, from 14:14 to 15:14, produced a cumulative loss of 11.2%. By 13:01 on October 20, cumulative drawdown from the pre-crash peak reached 28.6%. These figures are more consistent with disorderly liquidation than with an orderly correction.

The distinction matters. A falling market can still function. What the data show here is a market under extreme directional pressure, with rising volatility and deepening drawdown consistent with forced selling overwhelming near-term liquidity.

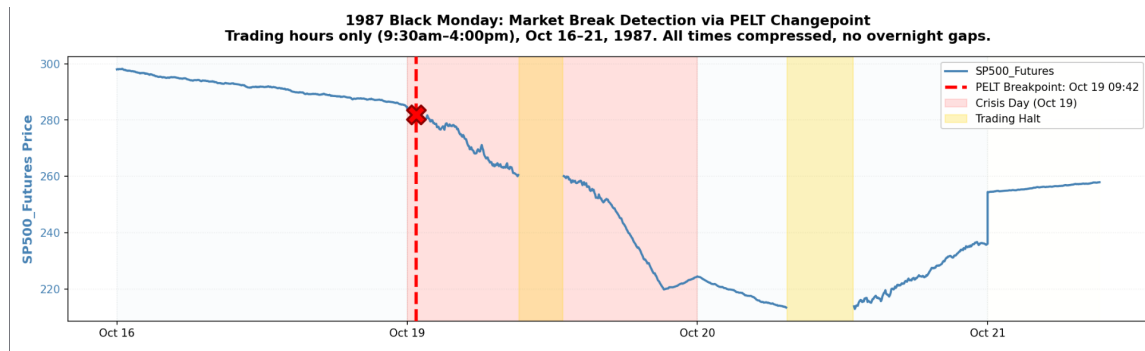


Figure 1: **1987 Black Monday: Market Break Detection via PELT Change-point.** S&P 500 futures prices (Oct 16–21, 1987) with PELT structural break identified at 09:42 on October 19. Red dashed line marks the changepoint; shaded regions indicate Crisis Day (Oct 19, pink) and Trading Halts (yellow). Trading hours only (09:30–16:00); no overnight gaps. The sharp discontinuity at 09:42 signals the threshold beyond which selling pressure becomes irreversible and liquidity begins to fail. *Source: 1987_crash_market_data.csv.*

HOW PORTFOLIO INSURANCE WORKED

Portfolio insurance was designed to replicate downside protection without requiring investors to buy a traditional put option. Instead of paying an upfront premium, institutions used dynamic hedging to reduce market exposure as prices fell, most often through sales of S&P 500 futures (Presidential Task Force on Market Mechanisms, 1988). Falling prices triggered cuts to exposure; rising prices restored it.

The logic was straightforward: if portfolio value falls, sell enough to limit further downside. The problem was that the hedge had to be adjusted continuously. It was not a static allocation; it depended on ongoing execution in live markets. That made the strategy highly sensitive to trading conditions.

In theory, dynamic hedging can work in a liquid and orderly market. In practice, the strategy assumed that large sell orders could be executed at or near quoted prices without materially moving the market—that liquidity would hold at scale. On October 19, that assumption failed almost immediately as market conditions deteriorated.

WHY THE ALGORITHMS SOLD

The algorithms sold because that is what they were built to do. They were not designed to identify undervaluation or buy into weakness—they were designed to reduce exposure as prices fell. The models did not malfunction. They followed their rule set under stress exactly as intended.

The issue was scale and synchronization. By late 1987, portfolio insurance strategies covered an estimated \$60–90 billion in assets (Carlson, 2007). Once the market declined

sharply on October 19, a large share of that capital faced the same imperative: sell futures to hedge additional downside. The result was concentrated, largely price-insensitive execution into a market already under pressure.

The effect did not remain confined to the futures market. As futures prices weakened relative to cash equities, index arbitrage transmitted that pressure into the stock market. CFTC data underscore the magnitude: gross futures selling by institutional hedgers increased from roughly 3,500 contracts on October 14 to 32,700 on October 19 (CFTC, 1988). Portfolio insurers represented a substantial share of non-market-maker futures selling on Black Monday (Carlson, 2006). That concentrated selling deepened the decline precisely when liquidity was thinnest.

This is the core strategic lesson. Portfolio insurance did not cause the crash by itself, but it amplified it through feedback. Falling prices triggered mechanical selling; that selling pushed prices lower; lower prices generated still more selling (Shiller, 1988). As Greenspan later observed, even if one investor could exit successfully, the system as a whole cannot all get out at once (Greenspan, 1988). The strategy did not fail because the models deviated from their design. It failed because market liquidity was insufficient to absorb the volume and synchrony of the required trades.

This diagnosis points to a single imperative: risk management systems must be redesigned so that portfolio-level discipline does not become market-level destruction. The 1987 and 2018 episodes reveal that mechanical risk controls, when operating at scale without awareness of liquidity, can amplify the very distress they are meant to contain. The following framework addresses this structural vulnerability.

RISK-CONTROL ARCHITECTURE

The Problem: Why Risk Management Becomes Destabilizing

Portfolio insurance fails catastrophically not because it identifies too much risk, but because it forces mechanical selling into collapsing liquidity. In 1987, algorithms required additional selling exactly as market conditions deteriorated—turning a circuit breaker into a system amplifier (Carlson, 2007; Fortune, 1993). The pattern repeated in 2018’s “Volmageddon”: coordinated mechanical rebalancing during stress amplified volatility, forcing mass liquidations at the worst possible prices (Augustin et al., 2021).

The root cause is simple: *discrete, price-triggered exits have no awareness of market liquidity and create incentives for synchronized selling across portfolios*. When multiple investors hit the same stop-loss levels simultaneously, execution quality collapses and forced selling becomes self-fulfilling. The investor intended to manage tail risk but instead becomes an unwitting agent of market instability.

Proposed Solution: A Liquidity-Aware, Three-Layer Architecture

We propose replacing reactive, trigger-based risk controls with a proactive system that phases de-risking across three layers, each designed to respect market capacity and conditions.

Layer 1: Volatility-Scaled Position Sizing (Continuous)

Primary control mechanism. As volatility rises, position leverage automatically declines, reducing portfolio delta before stress reaches critical levels (Moreira & Muir, 2017). This works continuously—not as a binary trigger—so the portfolio gradually shrinks its footprint as variance expands.

- **Rationale:** Markets can absorb gradual scaling; they cannot absorb synchronized liquidation.
- **Implementation:** Tie leverage multipliers inversely to realized and forecasted volatility.
- **Cost:** Whipsaw risk if markets rebound quickly after volatility spikes.

Layer 2: Liquidity-Aware Execution Limits (Conditional)

Operational safeguard. To prevent crowding and system risk, all de-risking trades must respect strict volume and spread thresholds: trade sizes capped at a percentage of recent average daily volume (e.g., no single sale exceeds 2% of ADV), and automatic pause rules triggered by extreme bid-ask spreads or failing auction mechanics.

- **Critical trade-off:** Execution may lag when liquidity evaporates. If the market gaps overnight, Layer 1 and Layer 2 cannot fully de-risk the portfolio in real time.
- **Rationale:** Preventing algorithmic crowding during stress is more important than achieving perfect fill prices (Fortune, 1993). A 1% worse entry into an illiquid market beats a 10% liquidation at any price.

Layer 3: Volatility-Conditioned Exit Thresholds (Backstop Only)

Final circuit breaker. This layer engages *only after* Layers 1 and 2 have already scaled down base exposure. Its purpose is not to drive routine de-risking, but to prevent unbounded tail losses if Layers 1 and 2 fail due to gap risk or extreme illiquidity. Dynamic thresholds reflect market conditions:

- Thresholds **widen** during high volatility (e.g., 8% loss level if $VIX > 40$) to avoid rigid, clustered exits that amplify market stress.

- Thresholds **tighten** in calm markets (e.g., 3% loss level if $VIX < 15$) where execution is orderly and losses are less likely to cascade.

Critical distinction: Layer 3 is *not* a parallel safeguard; it is sequential. By the time Layer 3 triggers, the portfolio has already been through Layers 1 and 2. This prevents the 1987 pattern where a single rigid rule fires at the worst moment and accelerates collapse.

What This Solves

1. **Prevents forced liquidation cascades.** Gradual scaling (Layer 1) avoids the 1987/2018 syndrome where synchronized selling destroys liquidity. Machines sell gradually; humans panic all at once.
2. **Respects market capacity.** Execution limits (Layer 2) prevent algorithmic crowding and shield the portfolio from being forced to execute into irrational prices during temporary liquidity voids.
3. **Breaks the feedback loop.** By reducing exposure early (Layer 1), the portfolio enters stress already lighter. Layers 2 and 3 protect what remains, ensuring that tail losses do not trigger cascading forced sales that destroy remaining portfolio value.
4. **Maintains control hierarchy.** Stop-loss rules are *defensive backstops*, not primary policy. They trigger only after proactive scaling and execution discipline have already been deployed. This prevents the rigid, mechanical exits that characterized portfolio insurance in 1987.

Residual Risks & Trade-Offs

1. **Whipsaw losses:** Selling into temporary volatility spikes followed by quick reversals will underperform a buy-and-hold approach in volatile but ultimately stable regimes. This is the deliberate cost of reducing tail risk.
2. **Execution quality degradation:** Pause rules and volume caps mean de-risking may be slow when speed matters. The portfolio may endure temporary larger drawdowns than it would under normal market conditions.
3. **Overnight gap risk:** No dynamic framework prevents gaps or limit-down moves. This is unavoidable under extreme illiquidity; the goal is to reduce tail exposure sufficiently that gaps do not trigger bankruptcy.
4. **Calibration risk:** Volatility estimation methods, volume thresholds, and spread

limits must be market-specific and regime-dependent. Miscalibration can lead to over-scaling or false confidence.

Implementation & Transferability

These principles are conceptually universal—all modern financial systems remain vulnerable to herding and structural instability (Haldane & May, 2011). However, **execution is market-specific**: volatility estimation methods vary by asset class, liquidity thresholds must reflect venue fragmentation, and execution limits depend on typical bid-ask spreads and ADV.

Before deployment, each market application requires calibration against historical stress episodes (2008, 2015, 2018, 2020) to validate that Layer 1 and Layer 2 would have scaled down exposure sufficiently to avoid catastrophic Layer 3 breaches.

Bottom Line

The fundamental shift is from “*act when price falls to X*” to “*continuously scale inversely to stress, execute respectfully, and only exit if all else fails.*” This prevents risk management from becoming a source of systemic instability while preserving the portfolio’s ability to contain tail losses. No framework eliminates all losses or gap risk, but this architecture materially reduces the probability of forced liquidation during the exact moments when market capacity is most depleted.

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